

7. Virtualization

Type 2 hypervisor is less efficient and used for personal use.

Bare Metal: Type 1 hypervisor is installed directly on top of the hardware, so no resources are wasted on the base os. They are used more in the enterprise environment, they are more efficient.

Ex: VMware ESXi, Microsoft Hyper-V, Proxmox

Advantages of virtual machines:

testing out software programs for different os

running something that could be a malware

learning about different cybersecurity concepts

ease and speed - (agility) to use the resources and unallocate them after use

saving cost for more computers to run different os and different activities

spin the resources down or reprovision if needed

snapshots for backups

Containers are there to scale up for the resources, containers are there to virtualize an application, not the entire container.

install virtual box and the extension pack, check if it is still supported.